SHoEbox: unified regional learning

2026-11-07 | Continuation of the regional-recall milestone

The ordinary seven-node regional graph now learns when to publish, retain state, and adapt retained payloads. Natural language and memory tasks use the same regional experts, byte interface, recall slots and learning path. Downstream read demand becomes a trainable input to the next round's publication decisions.

Completed learning run

Selected model	Update	Language NLL	Answer NLL	Exact NLL	Long exact NLL
Delayed credit	160	3.4339	2.1084	1.8644	1.9450
No delayed credit	160	3.5689	2.5569	2.2291	2.2378

Lower NLL is better. Language uses 1,024 held-out byte targets; task test uses 80 answer targets in 8 fresh documents. Long-delay documents triple the configured delays. Exact answers are a subset of the combined answer score. The credit arm's exact-answer accuracy is 40.7% on the ordinary test and 10.3% on the long-delay test.

The credit arm's validation language NLL moved from 4.2689 to 3.4429; answer NLL moved from 5.7476 to 2.1024. At the selected checkpoints, adding delayed credit changed test language NLL by -0.1350 and answer NLL by -0.4485 versus the matched arm. This comparison jointly switches delayed gate and payload gradients; live-window gradients remain enabled in both arms.

Continue tracking language and answer metrics separately as mixed-data exposure increases.

What is integrated

- A/B/C/U/V learn independent write-or-hold policies from candidate content, stored content and preceding-round demand. P/Q continue publishing readout outputs.
- Holds preserve publication contents, identity and provenance. Optional state holds also preserve contextualizer and recurrent contents; execution clocks advance.
- Receiver demand aggregates mean and peak soft selection probability across valid readers. It cannot affect a writer earlier in the same round.
- Bounded owner/version/epoch-bound sensitivities carry conditional gate and low-rank payload credit across training cuts. No replay buffer is introduced.
- Language and memory sources share all trainable weights but retain separate runtime state, demand, credit and stream cursors.

Matched exposure per arm

Item	Value
Parameters	312,440 (729 added to the parent)
Regional nodes / rounds / recall slots per receiver	7 / 2 / 7
Updates / batch / byte window / gradient segment	160 / 4 / 32 / 16
Natural-text / procedural-memory bytes	10,240 / 10,240
Training answer targets	624
Expert executions, retained trajectory	3,440,640
Additional inference demand state per stream	40 bytes
Training eligibility per stream	8,128 bytes

Both arms start from the same preceding graph weights, with fresh optimizer and runtime state. All weights train. AdamW uses learning rate 0.002, constant through the run. Selection minimizes the mean of validation language and answer NLL at updates 0, 40, 80, 120 and 160. Pauses add no selection opportunities.

Whole-rollout controls

These changes start from fresh test streams and include effects on later writes and state. Positive changes mean the intervention worsened NLL.

Control on selected credit model	Language delta	Answer delta
Block regional reads	+0.3813	+1.0858
Remove demand feedback	+0.0024	-0.0001
Always commit producer state	-0.0049	+0.0004
Always publish	+0.0032	+0.0028
Pool recall roles	+0.0045	-0.0004
Remove both feedback paths	+0.0043	+0.0002

Learned publication behavior

Owner	Writes per execution	Mean demand	Peak demand
A	83.6%	0.395	0.709
B	85.5%	0.567	0.950
C	69.9%	0.373	0.843
U	82.4%	0.333	0.514
V	99.6%	0.211	0.420

Retention and next steps

At answer queries, selected valid publications averaged 0.18 bytes old; 0.0% were older than the 16-byte gradient segment. These measurements cover selected edge reads across all receivers and rounds; they do not assign a stored publication to a particular fact.

The learned demand projection norm is 0.0921; the memory-feature norm is 0.4385; the payload-adapter coefficient norm is 0.0768. The training archive records delayed-gradient norms separately for both sources.

Verification and interpretation

The full five-suite run passed 543 checks. Targeted numerical and evaluation-cache checks bring the total to 547 unique checks. All 19 mixed/graph regressions passed after the clipping update, followed by the evaluation-cache check. Fourteen new integration checks cover causal demand, exact hard holds, bootstrap and resets, multi-round recording, owner/version/epoch rejection, finite task gradients, conditional gradient equivalence, single-use receipts, legacy behavior, source isolation, and exact optimizer/state/credit/cursor resume. Evaluation checkpoints each finished measurement and reuses results for identical weights.

The delayed estimator holds historical candidates, feature/demand networks and cross-region candidate histories fixed. It restores the documented conditional publication derivative; it does not reconstruct the full recurrent history. Gate eligibility decays by 0.995 per active regional round.

Training also gained overflow-safe gradient clipping. It retains the preceding FP32 arithmetic whenever the norm is finite, otherwise checks every gradient entry and computes the norm in FP64 before scaling. Actual nonfinite entries still stop the update. Recovery counts are included in the checkpoint totals. The control resumed from update 120 after the original interruption; discarded work is excluded from the retained-trajectory counts and adds no selection opportunity.

The architectural gap is closed: the general graph can now train useful publication behavior with task feedback. The next pass should balance language and answer learning, increase mixed-data exposure and inspect selective retention under broader language contexts. Keep the switches for state commits, demand, role composition and delayed credit. Code comments explicitly preserve the intended coordination as a potentially threshold-dependent emergent feature for later ablation.

Reproduce and inspect

The archive contains source, the exact corpus and parent, selected/final checkpoints, optimizer/runtime/credit states, source cursors, query traces, metrics, complete test output and a hash manifest.

```
cd SHoEbox
python scripts/run_graph_publication.py --parent ../graph_publication/parent.pt \
  --data data/tiny_shakespeare.txt --out artifacts/graph_publication --parallel
python scripts/check_all.py
python scripts/summarize_graph_publication.py
```

Architecture and controls: `docs/graph_publication.md`. Machine-readable results: `experiments/text_regions/results/graph_publication/summary.json`.